

KIRKPATRICK'S LEVELS OF TRAINING CRITERIA: THIRTY YEARS LATER

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Kirkpatrick's model (1959a, 1959b, 1960a, 1960b) of training evaluation criteria has had widespread and enduring popularity. This model proposed four "levels" of training evaluation criteria: reactions, learning, behavior, and results. Three problematic assumptions of the model may be identified: (1) The levels are arranged in ascending order of information provided. (2) The levels are causally linked. (3) The levels are positively intercorrelated. This article examines the validity of these assumptions, the frequency of each level in published evaluation studies, correlations from the literature in regard to Assumptions 2 and 3, and implications for the researcher and training manager.

About 30 years ago, Kirkpatrick published a series of articles in which he outlined four categories of measures of the effectiveness of training outcomes (Kirkpatrick, 1959a, 1959b, 1960a, 1960b). Each category was termed a "step." Step 1 was termed *reactions* and defined as trainees' "liking of" and "feelings for" a training program. Note that a reaction measure is conceived in attitudinal rather than behavioral terms. Step 2, *learning*, was defined as "principles, facts, and techniques understood and absorbed" by the trainees. Step 3 was *behavior*, defined as "using [learned principles and techniques] on the job". Step 4, *results*, was spoken of simply as the ends, goals, or "results desired . . . reduction of costs; reduction of turnover and absenteeism; reduction of grievances; increase in quality and quantity of production; or improved morale."

This proposed model or taxonomy of training evaluation criteria clearly met a felt organizational need for it quickly became well known in training departments around the country. Moreover, the field of industrial/organizational psychology has largely accepted this framework (Cascio, 1987).

The power of Kirkpatrick's model is its simplicity and its ability to help people think about training evaluation criteria. In other words, it provides a vocabulary and rough taxonomy for criteria. At the same time, Kirkpatrick's model, through its easily adopted vocabulary and a number

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of (often implicit) assumptions, can tend to misunderstandings and over-generalizations. We discuss the model in the light of three assumptions that appear to be largely implicit in the minds of researchers and trainers, although to all appearances unintended by Kirkpatrick himself when the model was proposed. These three assumptions, however, have been endorsed or can be identified in the literature.

The first assumption is that the "steps" are arranged in ascending value of information provided (Newstrom, 1978). A measure of learning provides more information than does a measure of reaction, and so forth. It is, in fact, now common to see the term "levels" of criteria referred to instead of the more purely procedural term "steps" (Goldstein, 1986a; we use the term "levels" in the remainder of the article). The second assumption is that these levels of evaluation are causally linked. For example, Hamblin (1974) states, "training leads to reactions which lead to learning which leads to changes in job behavior which lead to changes in the organization." A third assumption is that the levels are positively intercorrelated. That is, a set of essentially positive interrelationships, or "positive manifold," is posited to exist among levels of training evaluation (cf. Newstrom, 1978). Each of these three assumptions about Kirkpatrick's steps appears to be codified in what has been termed the "hierarchical model" of training evaluation (Hamblin, 1974; Noe & Schmitt, 1986), where "favorable outcomes at the lowest criterion level are seen to be necessary for favorable outcomes to occur at the next higher level, and so on" (Clement, 1982).

Examining the Assumptions

It is our belief that each of these assumptions about levels of training evaluation is problematic. We address each theoretically and then discuss a literature overview, including relevant correlations from the literature to further examine the third assumption.

Assumption 1: Each succeeding level is more informative than the last.

This assumption has the flavor of reasonableness, which may explain a nearly absolute lack of discussion about it in the literature. There are, however, some questions that can be raised in regard to it. First, it is not clear that all training in organizations is meant to effect change at all four levels. Some training may be largely rewarding, spirit-building, or perquisite in nature. For example, programs designed to instill company pride or rejuvenate employees may be realistically expected to have impacts at the reaction level only. Similarly, training that is limited to inculcation of company history or philosophy may, in some cases, be best and most

appropriately measured by growth in knowledge: the goal may be a knowledgeable manager, and attendant behavior change, for example, may not be of immediate interest to the evaluator.

A point worth making in this regard is that Assumption 1 can lead to a perception of Level 4, results, as invariably the "best" measure, since it is highest in the hierarchy. Advocacy of the "dollar criterion" (Brogden & Taylor, 1950), having been championed in research journals (e.g., Reilly & Smither, 1985; Weekly, Frank, O'Connor, & Peters, 1985), now appears in popular practitioner journals (e.g., Fitz-enz, 1985; Sheppeck & Cohen, 1985) and management textbooks (e.g., Hall & Goodale, 1986). Such sources promise that techniques for dollar estimation of intervention effectiveness will help HRM managers to "strengthen their power and influence" (Hall & Goodale, 1986). Estimating intervention value in dollars is thus enthusiastically presented as one way to emerge from the dark ages of the "personnel" function and move into the new era of the "human resources" department, which is (in theory) happily revenue producing instead of resource draining. Nonetheless, even when practically possible, there may be times when dollar estimation is inappropriate, as when the plant manager argued against quantitative evaluation because it was important to keep the idea of the intervention in the fore (Saporito, 1986). That there is a tendency for the purely quantitative, as dollars surely are, to obscure the field of possible approaches to issues such as evaluation is attested to by Goldstein (1986b, p. 22):

It is startling to discover that most I/O psychologists still believe that quantitative models are good while qualitative research models do not contribute. . . . [But] the continuum from good to bad does not have anchors with quantitative on the good side and qualitative on the bad side.

Assumption 2: Each level is caused by the previous level.

Causality is difficult to prove or disprove. Several questions can be raised about this assumption, however. The first question is one of temporality. While Level 3 and 4 measures occur at some time after training, Levels 1 and 2 usually are administered directly after training (as well as possibly before, in the case of Level 2). Indeed, a single instrument may measure both reactions and learning (Alliger & Horowitz, 1989). There would seem, then, to be no temporal distinction between reactions and learning as far as their assessment. Why then assume the former causes the latter? Probably the rationale is that those students reacting well to training are attending to the training as well. But Lewicki's (1986) studies, for example, indicate that attending can be deleterious to learning; other psychologists have argued that people are not good at reporting their experiences of learning (Hofstadter & Dennett, 1981; Thorndike, 1935).

Moreover, perhaps it is only when trainees are challenged to the point of experiencing the training as somewhat unpleasant that they learn: In such a case learning and reactions might be *negatively* correlated. Such negative correlations between Level 1 and Level 2 measures have in fact been found in educational classroom research (Remmers, Martin, & Elliott, 1949; Rodin & Rodin, 1972). Or consider that humorous lectures are liked better (Level 1) but do not cause more learning (Level 2) (Kaplan & Pascoe, 1977).

There is some reason to believe, then, that reactions may not be expected to cause learning. (This is of course more likely to the extent that reaction measures are attitudinally and not behaviorally based.) At the same time, among learning, behavior, and results, some causal links should exist. Learning often should relate to behavior since, in some cases, grasp of some principle, sequence, or facts can be important to behavior. Level 3, behaviors, should cause Level 4, results. In a reverse causality, Level 4, results, should be important to the maintenance of Level 3, behaviors, since people will tend to continue behaviors that are perceived to be effective. That is, feedback sustains the behavior-results link. Figure 1 illustrates both the traditional hierarchical causal model (Clement, 1982) and this alternative proposed causal linking among levels of criteria, where Level 1 is unrelated to the other levels, Level 2 to some degree is important in the causality of Levels 3 and 4, and Levels 3 and 4 are causally interdependent.

Assumption 3: Each succeeding level is correlated with the previous level; or, more generally, there exists a "positive manifold": all correlations among levels are positive.

This assumption is clearly linked to the previous one. If each level is causally linked to the previous level, then positive correlations among measures should exist. But, considering the points already made about causal linking among levels, to what extent, is it in fact logical to assume this set of positive relationships? For example, it has been argued that perhaps no positive relationship should be predicted between reactions and learning. On the other hand, our theoretical expectations are that Levels 2, 3, and 4 would be intercorrelated.

To some degree, evidence against the validity of Assumption 3 is seen in the literature. Many evaluation studies that have evaluated training on two or more of Kirkpatrick's levels have reported different effects of training for different levels (e.g., Campion & Campion, 1987; Decker, 1982; Meyer & Raich, 1983; Moffie, Calhoun, & O'Brien, 1964; Russell, Wexley, & Hunter, 1984). This provides indirect evidence against the likelihood of finding high positive intercorrelations among levels. An examination of

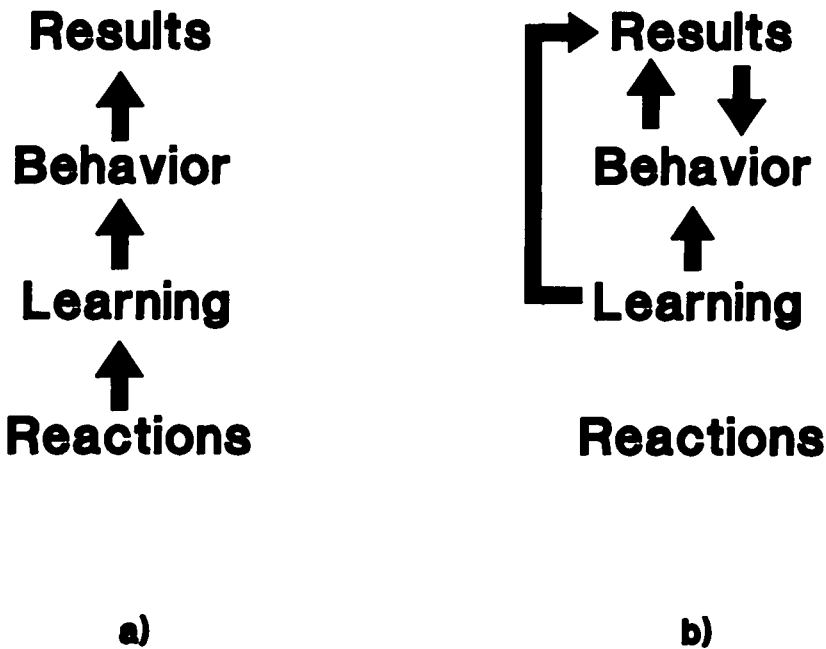


Figure 1: The a) causality in the hierarchical model and b) an alternative model of causality among Kirkpatrick's four levels of training criteria

correlations among levels, which provide a more direct examination of Assumption 3, follows.

An Examination of the Intercorrelation Among Levels

One way to examine empirically the accuracy of the above assumptions is to review the literature for reported correlations among criteria, categorized by Kirkpatrick's levels. Accordingly, a citation search was conducted on Kirkpatrick (1959a, 1959b, 1960a, 1960b, 1967, 1976, 1977, 1978, 1979, 1985) for the years 1969 to the present (the Social Science Citation Index began in 1969). Of the 55 articles so located, only 8 dealt with the actual evaluation of a training program; none reported correlations between levels. Consequently, a literature search was conducted for any article dealing with evaluation of a training program in several journals that might be expected to publish such articles. Journals were reviewed

TABLE 1
Breakdown of Literature Review by Journal and Level of Criteria

	Journal						Total
	AMJ	JABS	JAP	PP	P	TDJ	
Years reviewed	59-88	65-88	59-88	59-88	59-88	60-88	
No. articles reporting training evaluation results	18	41	79	21	10	32	203
Criteria levels used							
1	1	1	2	-	1	6	11
2	8	9	29	9	2	4	61
3	5	16	34	4	2	5	66
4	-	5	3	-	2	1	11
1, 2	1	1	6	3	-	2	13
1, 3	-	6	-	1	1	10	18
1, 4	-	-	-	1	-	1	2
2, 3	1	1	4	1	-	1	8
2, 4	-	-	-	-	-	-	0
3, 4	1	1	-	-	1	-	3
1, 2, 3	1	-	1	-	1	-	3
1, 2, 4	-	-	-	-	-	-	0
1, 3, 4	-	1	-	-	-	1	2
2, 3, 4	-	-	-	-	-	-	0
1, 2, 3, 4	-	-	-	2	-	1	3
Design							
Field	14	33	38	19	10	30	134
Laboratory	4	8	41	2	0	2	57

Note: AMJ = *Academy of Management Journal*, JABS = *Journal of Applied Behavioral Science*, JAP = *Journal of Applied Psychology*, PP = *Personnel Psychology*, P = *Personnel*, TDJ = *Training and Development Journal*.

from 1959 to present, if possible. Specifically, we reviewed *Academy of Management Journal*, *Academy of Management Review* (first published in 1976), *Journal of Applied Behavioral Science*, *Journal of Applied Psychology*, *Personnel Psychology*, *Personnel*, and *Training and Development Journal* (*Journal of ASTD* until 1966). Table 1 shows the number of articles found by journal and category. Only one relevant article was found in *Academy of Management Review*; it did not report criteria, and so that journal is not included in Table 1.

Table 1 illustrates some interesting facts. First, most articles in the journals reviewed look at a single level of evaluation. Second, in contrast to the results of Catalanello & Kirkpatrick's (1968) and Kirkpatrick's (1978) industry surveys, most studies do not focus on reactions alone. In all probability this reflects the difference between practice and publishing: in practice, most training is evaluated on the reaction level only, while editors for the reviewed journals may look for evaluation results measuring learning, behavior, or results. Third, field studies are well represented (70% of total). It should also be noted that our search turned up numerous articles that addressed training evaluation but that did not report an evaluation study. For example, *Training and Development Journal* actually published

in this time period 214 articles on training evaluation; only the 32 shown in Table 1 reported an evaluation study.

Unfortunately, of the articles in Table 1, only eight reported intercorrelations among two or more levels of evaluation. Consequently, we widened our search again by following leads from reference lists in these articles and appropriate other articles and books. In the end, we found 16 articles reporting 30 appropriate correlations. Four of these articles were academic samples relating teacher ratings and student evaluations; our interest was in industrial training. Consequently, our final data set consisted of only 12 articles reporting 26 correlations. Although disappointed in this small number of usable results, we felt our search was relatively successful, given the apparent sparsity of correlational data of this type: Clement (1982) reported being unable to locate in the literature *any* correlations between levels of Kirkpatrick's model. Actually, the fact that the vast majority of articles reviewed did not report inter-level correlations, even when it was highly likely the data would have allowed such correlations to be calculated, may signify that the hierarchical model is simply assumed to be correct. Table 2 provides a list of the 11 articles, and information about the samples studied. Table 3 provides the correlations between levels of evaluation, listed in descending magnitude. Sample size-weighted mean correlations for each pair of levels are found in parentheses.

Results and Discussion

As can be seen, the number of available correlations between pairs of levels ranged from one to eight. This small number of correlations means, obviously, that generalizations should be made with care. An examination of the mean sample size-weighted correlations for each cell in Table 3 indicates that Level 1 may tend to correlate only slightly with the other levels: $r_{1,2} = .07$; $r_{1,3} = .05$; and $r_{1,4} = .48$ (but this last is based on a single correlation). On the other hand, the weighted mean correlations among Levels 2, 3, and 4 could be said to be slightly larger: $r_{2,3} = .13$; $r_{2,4} = .40$; and $r_{3,4} = .19$. Is it in fact possible that attitudinal reactions to training should be considered in a category independent from such constructs as learning or behavior, as these mean correlations might suggest and as is depicted in Figure 1? The data from our literature search certainly do not warrant a definite answer to this question. For example, it is certainly possible that some of these correlations are artifactually small, due to the reduced variance typical of reaction measures (cf. Alliger & Horowitz, 1989). The variance of the reaction measures was available for only five of the articles listed in Table 2; when we examined the variance relative to the means for those measures, reduced reaction measure variance could be concluded as probable in some cases but not in others. In any

TABLE 2
Articles Reporting Correlations Between Levels of Evaluation

Author(s)	N	Type of sample	Description of measures (Level: Type)
Alliger	90	1st- to 5th-level managers	2: Posttest 3: Questionnaire
Alliger & Horowitz	1,259	1st-level managers	1: Questionnaire 2: Pre posttest
Bolman	118	Workers (type unspecified)	1: Questionnaire 2: Self-ratings
Clement	50	1st-level supervisors	1: Questionnaire 2: Pre posttest 3: Self-ratings 4: Subordinate ratings
Eden & Shani	105	Male soldiers	1: Questionnaire 2: Posttest
Miles	34	School Principals	2: Observer ratings 3: Peer ratings
Noe & Schmitt	60	School Administrators	1: Questionnaire 2: Observer ratings 3: Observer ratings 4: Supervisor ratings
Reeves & Jensen	173	Workers (type unspecified)	1: Questionnaire 3: Self-ratings
Severin	50 ^a	_____ ^a	2: Training grades 3: Supervisor ratings 4: Production records
Smith	60	Branch office managers	2: Pre posttest 4: Customer satisfaction
Stroud	103	Supervisors	1: Questionnaire 3: Self-ratings Supervisor ratings
Wexley & Baldwin	256	Management students	1: Questionnaire 2: Posttest 3: Self ratings Observer ratings

^aSeverin's article was itself a compendium of many studies; the median N was 50, and the nature of the samples varied.

case, the proposed model of Figure 1 may perhaps stand as an alternative that, even if ultimately incorrect, may stimulate further research. Such research would seem useful, since practitioners and researchers in training are often faced with decisions regarding the nature of training evaluation data.

Clement (1982), has stated that correlations that do not support the hierarchical model fail to do so because of noise from intervening variables such as motivation, context of transfer, trainee attitudes, and so forth. No doubt these variables account for some of the variation among correlations in Table 3.

In this regard, one question that might be raised from Assumptions 2 and 3 is whether all correlations in Table 3 may be seen as arising from a

TABLE 3
Correlations Between Levels of Evaluation

	Level 2	Level 3	Level 4
Level 1	.79 ^e .50 ^d .35 ^c .17 ^g .07 ^f -.03 ^b (.03)	.12 ^k .04 ⁱ .06 ^g .01 ^h (.04)	.48 ^g
Level 2		.74 ^j .40 ^d .16 ^a .06 ^f .06 ⁱ -.02 ^d -.05 ^d -.15 ^g (.18)	.53 ^g .24 ⁱ (.40)
Level 3			.48 ⁱ .41 ^g .20 ^d .00 ^d -.07 ^d (.19)

Note: For Noe and Schmitt data, original study correlations determined by adding residual to reproduced correlations; for Reeves Jenkins, an average r is presented; for Severin, and Wexley Baldwin, median correlations are reported; for Stroud, correlation in table represents midpoint of range reported (.036 to .211). Mean sample size-weighted correlations for each cell are in parentheses.

^aAlliger (1988); ^bAlliger & Horowitz (1989); ^cBolman (1971); ^dClement (1982); ^eEden & Shani (1982); ^fMiles (1965); ^gNoe & Schmitt (1986); ^hReeves & Jensen; ⁱSeverin (1952); ^jSmith (1976); ^kStroud (1959); ^lWexley & Baldwin (1986).

single distribution, sampling error accounting for differences among correlations. That is, if all levels are intercorrelated to about the same degree, then sampling error might be the only source of variance among correlations within and among levels. However, predicted variance due to sampling error accounts for less than 15% of the variance of the distribution consisting of all correlations in Table 3.

An issue of interest not addressed in this article is the relationship among criteria *on the same level*. Should researchers be satisfied that a single criterion on a given level will completely capture that level, or that other criteria on that level would have told the same story about training effectiveness? We would like to suggest that caution is needed here: the measurement of different criteria on the same level could show different results. Consider, for example, that behavior learned in a training class might be differentially displayed to, and evaluated by, superiors, peers, or subordinates.

Kirkpatrick's model may never have been meant to be more than a first, global heuristic for training evaluation. As such it has done well.

There are, in fact, several other models of training evaluation criteria, most of them very similar to Kirkpatrick's (e.g., Jackson & Kulp, 1978; Warr, Bird, & Rackham, 1970). However, we have proposed that each of three assumptions apparently implicit in the general understanding of Kirkpatrick's model, and in that of most other models as well, can be logically questioned. We have also shown that much of the literature available underscores the problematic nature of these assumptions.

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